

Benchmarking Arabic Large Language Models: A Comprehensive Literature Review

Ruoyu Zhang

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1 Introduction

Large language models (LLMs) have transformed natural language processing (NLP), yet Arabic remains underrepresented in evaluation. Arabic presents unique challenges including diglossia (Modern Standard Arabic versus regional dialects), rich morphology, and cultural complexity. To address these issues, a wide array of benchmarks and leaderboards have been introduced between 2023 and 2025, spanning broad multi-task suites, cultural and dialectal evaluations, domain-specific tests, safety and trustworthiness benchmarks, retrieval-based tasks, embedding evaluations, and multimodal settings. This paper provides a comprehensive literature review, organized by benchmark categories, and highlights their strengths, limitations, and methodological implications.

2 Benchmark Categories

2.1 Broad Multi-Task and Reasoning Benchmarks

- **ORCA** (Elmadany et al., 2023): 60 datasets, seven NLU clusters. Strength: dialect-inclusive. Limitation: lacks generative evaluation.

- **LAraBench** (Abdelali et al., 2024): 33 NLP + speech tasks. Strength: gap analysis. Limitation: mixed modalities.
- **BALSAM** (Al-Matham et al., 2025): 78 tasks, 14 categories. Strength: unified, anti-leakage. Limitation: downplays sequence labeling.
- **Arabic MMLU** (Koto et al., 2024): Arabic-native multiple-choice exams. Strength: rigorous. Limitation: static and MCQ-only.
- **Qiyas** (Al-Khalifa & Al-Khalifa, 2024): Arabic aptitude test. Strength: standardized. Limitation: narrow scope.

2.2 Cultural, Dialectal, and Alignment Benchmarks

- **AraDiCE** (Mousi et al., 2025): Seven dialect datasets + cultural QA. Strength: dialect breadth. Limitation: MT + post-editing.
- **Palm** (Alwajih et al., 2025): Culturally inclusive dataset. Strength: pan-Arab coverage. Limitation: not standardized.
- **SaudiCulture** (Ayash et al., 2025): 441 questions across Saudi regions. Strength: regional detail. Limitation: Saudi-only.
- **CamelEval** (2024): 805 prompts for generative + cultural QA. Strength: cultural eval. Limitation: judge reliability.

2.3 Safety, Trust, and Ethics

- **AraTrust** (2024): Evaluates bias and hallucination. Strength: ethical evaluation. Limitation: narrow scope.
- **Ashraf et al.** (2025): 5,799 Arabic safety prompts. Strength: red-teaming. Limitation: dialectal safety gaps.

2.4 Domain-Specific Benchmarks

- **ArabLegalEval** (Hijazi et al., 2024): Legal multitask evaluation. Strength: domain specificity. Limitation: Saudi focus.
- **MedArabiQ** (Abu Daoud et al., 2025): Seven medical tasks. Strength: domain realism. Limitation: limited expert judging.
- **AraFinNLP / ArBanking77** (Hariri & Farha, 2024): 39k banking queries. Strength: large-scale. Limitation: intent-only.
- **AraSTEM** (Mustapha et al., 2025): 11.6k STEM MCQs. Strength: breadth. Limitation: MCQ-only.

2.5 QA and Retrieval Benchmarks

- **ArabicaQA** (Abdallah et al., 2024): 92k QA pairs. Strength: scale. Limitation: dialect gap.
- **AraDPR** (Abdallah et al., 2024): Dense retriever. Strength: first Arabic retriever. Limitation: initial release.
- **SILMA RAGQA** (2025): Retrieval-augmented bilingual QA. Strength: hybrid approach. Limitation: preliminary.

2.6 Religion and Society

- **Hajj-FQA** (Aleid & Azmi, 2025): 800 fatwas, 2.8k QA pairs. Strength: religious relevance. Limitation: narrow scope.
- **AraHealthQA** (2024): Mental health QA shared task. Strength: societal relevance. Limitation: small scale.

2.7 Embedding and Classical NLP

- **ArabicMTEB** (Bhatia et al., 2025): 94 datasets across 8 task types. Strength: scale. Limitation: similarity only.
- **Swan** (Bhatia et al., 2025): Arabic-centric embeddings. Strength: dialect-aware. Limitation: embedding-only.
- **Cross-lingual NER** (Al-Duwais et al., 2024): Explored transfer for NER. Strength: cross-lingual insights. Limitation: task-specific.

2.8 Multimodal Benchmarks

- **CAMEL-Bench** (Ghaboura et al., 2024): 29k multimodal Qs. Strength: large multimodal coverage. Limitation: MSA-only.
- **Peacock & Henna** (Alwajih et al., 2025): Arabic multimodal with cultural Henna dataset. Strength: cultural multimodality. Limitation: OCR weaknesses.

Benchmark	Category	Tasks	Method	Strengths	Limitations
ORCA	Broad NLP	NLU tasks	Fine-tuned, metrics	Dialect-inclusive	No generative tasks
LAraBench	Broad NLP	NLP + speech	Zero/few-shot	Gap analysis	Mixed focus
BALSAM	Broad NLP	78 tasks	Ref. + LLM-judge	Unified, anti-leakage	Limited tagging
Arabic MMLU	Reasoning	Exams	MCQ	Native rigor	Static
Qiyas	Reasoning	Aptitude	MCQ	Standardized	Narrow

AraDiCE	Culture	7 dialects + culture	Mixed eval	Dialect breadth	MT-derived
Palm	Culture	Pan-Arab	Mixed eval	Inclusivity	Not standardized
SaudiCulture	Culture	441 Qs	MCQ, open	Regional depth	Saudi-only
AraTrust	Safety	Bias/ethics	Mixed eval	Ethical dimension	Narrow
MedArabiQ	Domain	Medical QA	Acc, BERTScore, LLM-judge	Domain realism	Limited experts
ArabLegalEval	Domain	Legal QA	Acc/F1	Legal specificity	Narrow jurisdiction
AraFinNLP	Domain	Finance intents	Classification	Domain specificity	Limited scope
AraSTEM	Domain	STEM MCQs	MCQ	Large STEM coverage	MCQ-only
ArabicaQA	QA	92k Qs	Accuracy, retriever	Scale, human-annotated	Dialect gap
Hajj-FQA	Religion	Fatwa QA	Extractive/generative	Religious relevance	Narrow
ArabicMTEB	Embedding	94 datasets	Sim metrics	Largest embedding eval	Similarity only
CAMEL-Bench	Multimodal	29k multi-modal	Task-specific metrics	Largest multi-modal	MSA-only
Peacock/Henna	Multimodal	Cultural multi-modal	LLM-judge	Cultural multi-modality	OCR weak

3 Methodological Pros and Cons

- **MCQ likelihood:** Simple, reproducible. Cons: not generative. Use: knowledge recall.
- **Exact-match / F1:** Clear answers. Cons: penalizes paraphrases. Use: extractive QA, NER.
- **BLEU/ROUGE:** Standard metrics. Cons: weak human correlation. Use: auxiliary check.
- **LLM-as-judge:** Scalable, aligned. Cons: judge bias. Use: generative, cultural, safety tasks.
- **Human experts:** Reliable. Cons: costly, slow. Use: medicine, law.
- **Zero-shot prompting:** Tests pretraining. Cons: prompt-sensitive. Use: baseline.
- **Few-shot prompting:** Boosts performance. Cons: not standardized. Use: applied evaluation.
- **Fine-tuning:** High accuracy. Cons: expensive, domain-bound. Use: specialist tasks.
- **Blind test servers:** Prevent leakage. Cons: limited accessibility. Use: leaderboards.

4 Discussion and Gaps

The reviewed benchmarks collectively illustrate a rapid maturation of Arabic LLM evaluation, yet several important gaps remain.

First, **dialectal coverage is incomplete**. While AraDiCE, Palm, and SaudiCulture incorporate dialectal material, the majority of benchmarks remain MSA-centric. Dialects such as Maghrebi (e.g., Moroccan, Algerian, Tunisian), Iraqi, Yemeni, and Sudanese Arabic remain underrepresented, despite their wide use in everyday communication. The consequence is that models may appear strong on MSA benchmarks while failing in realistic user interactions that often blend dialect and code-switching.

Second, **code-switching evaluation is almost absent**. Many Arabic speakers combine Arabic with English or French within a single conversation. Current benchmarks do not systematically

assess how LLMs handle such hybrid language use, which risks misrepresenting their practical performance in real-world contexts.

Third, **interactive and multi-turn dialogue is rarely tested**. Most benchmarks use single-turn QA or classification formats. Few tasks assess consistency across multiple dialogue turns, memory of prior context, or style adaptation to user formality. This leaves a blind spot in evaluating Arabic LLMs as conversational agents, a common use case in deployment.

Fourth, **safety, bias, and ethical alignment remain underdeveloped**. Although Ashraf et al. (2025) and AraTrust introduced safety-oriented datasets, most focus on a limited set of harmful prompts. Dialectal and euphemistic expressions of hate speech, extremism, or misinformation are underexplored. Furthermore, benchmarks seldom evaluate socio-cultural harms, gender or sectarian bias, or the appropriateness of responses in sensitive religious and political contexts. These omissions carry direct risks for responsible AI deployment.

Fifth, **grounded factuality and hallucination detection are not sufficiently benchmarked**. ArabicaQA and SILMA RAGQA begin to address retrieval-augmented generation, but few studies systematically test whether Arabic LLMs correctly attribute sources or resist hallucination when grounded in Arabic corpora. This is critical for domains such as journalism, healthcare, and law, where factual consistency is paramount.

Sixth, **evaluator robustness requires more attention**. Many benchmarks now rely on LLM-as-a-judge to approximate human evaluation. While efficient, these methods introduce biases (verbosity, position, stylistic preference) and rely on judge models whose own cultural or domain competence may be limited. Without bias calibration and human validation, reported scores may misrepresent actual model quality.

Finally, **benchmark standardization and sustainability remain challenges**. The field is fragmented, with overlapping datasets and divergent metrics. Few benchmarks implement blind, rotating test sets as in BALSAM. Without such practices, the risk of data contamination grows as models are trained on ever larger corpora, many of which include benchmark test data.

Significance of Findings

The significance of these gaps is twofold. First, they reveal that current benchmarks may overestimate Arabic LLM performance by focusing on easier, MSA-based, single-turn tasks with limited cultural grounding. Second, they highlight strategic opportunities: developing new benchmarks that target underrepresented dialects, code-switching, safety stress-testing, interactive dialogue, and grounded factuality. By addressing these blind spots, future research can create evaluation frameworks that more accurately reflect real-world Arabic language use and ensure that LLMs deployed in Arabic-speaking regions are both effective and culturally aligned.

5 Conclusion

Arabic LLM benchmarking evolved from translated multiple-choice datasets to native, generative, and culturally grounded evaluations. Benchmarks now span broad NLP, cultural and dialectal inclusivity, safety, domain-specific reasoning, retrieval, embeddings, and multimodal tasks. Yet gaps remain in dialects, code-switching, safety, and interactive dialogue. Future benchmarks should adopt blind test sets, hybrid evaluation (LLM + human), and broader domain/dialect coverage.

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